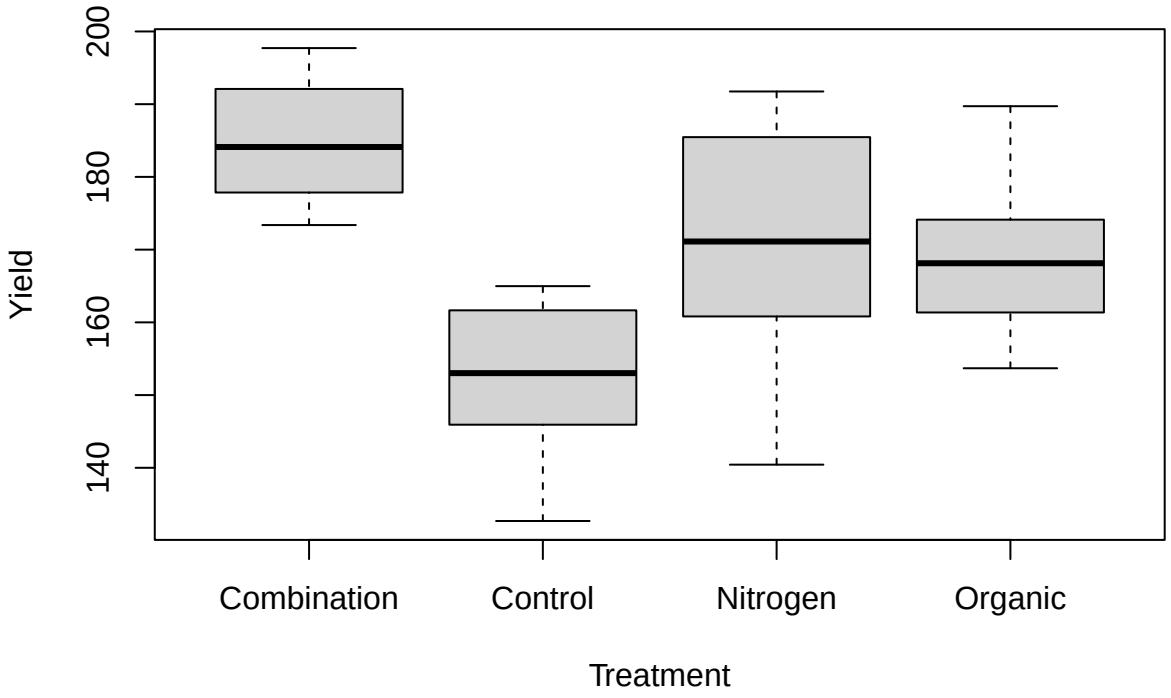
fertilizer-contrasts.R

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```
my_data <- read.csv('https://staturn-courses.github.io/stat-801a/notes/data/fertilizer-experiment.csv')
with(my_data, boxplot(Yield ~ Treatment))
```




```
# Fit the model
fit <- aov(Yield ~ Treatment, data = my_data)
anova(fit)
```

```
## Analysis of Variance Table
##
## Response: Yield
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Treatment  3 5496.1 1832.03  12.425 9.935e-06 ***
## Residuals 36 5308.2  147.45
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
emm <- emmeans(fit, ~Treatment)
emm
```

```
##  Treatment   emmean   SE df lower.CL upper.CL
##  Combination    184 3.84 36      177      192
##  Control        151 3.84 36      144      159
##  Nitrogen       170 3.84 36      162      178
##  Organic        169 3.84 36      161      177
##
## Confidence level used: 0.95
```

```r
# Contrasts
# 1. Fertilizer vs. no fertilizer
# 2. Nitrogen vs. organic
# 3. Combination vs. Nitrogen
# 4. Combination vs. Organic
# 5. Combination vs. avg of Nitrogen + Organic
contrast(emm, list(
  'Fertilizer vs. no fertilizer' = c(1,-3,1,1)/3,
  'Nitrogen vs. Organic' = c(0, 0, 1, -1),
  'Combination vs. Nitrogen' = c(1,0,-1,0),
  'Combination vs. Organic' = c(1,0,0,-1),
  'Combination vs. N + O' = c(2,0,-1,-1)/2
))
```

```
##  contrast                    estimate   SE df t.ratio p.value
##  Fertilizer vs. no fertilizer    23.10 4.43 36   5.209 <0.0001
##  Nitrogen vs. Organic             0.86 5.43 36   0.158  0.8750
##  Combination vs. Nitrogen        14.53 5.43 36   2.675  0.0112
##  Combination vs. Organic         15.39 5.43 36   2.834  0.0075
##  Combination vs. N + O           14.96 4.70 36   3.181  0.0030
```